Persistent, Fault-Tolerant UAV-Based IoT Networks

MARL-Based Swarm Reconfiguration

Team – Intelligent UAV Deployment

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Problem at a Glance

N **drones** fly over M **ground sensors**.

Each drone has a **finite battery** and can **fail**.

Every sensor needs **at least** K **drones** above it **at all times**.

Challenge: coordinate the swarm with *no central controller*.
Solution: each drone learns its own policy via multi-agent RL.

MDP Formulation

We model the system as a cooperative Dec-POMDP:

$$\mathcal{G} = \langle \underbrace{N}_{\text{agents}}, \underbrace{\mathcal{S}}_{\text{state}}, \underbrace{\{\mathcal{A}_i\}}_{\text{actions}}, \underbrace{\{\mathcal{O}_i\}}_{\text{obs.}}, \underbrace{\mathcal{T}}_{\text{dynamics}}, \underbrace{R}_{\text{reward}}, \underbrace{\gamma}_{\text{discount}} \rangle$$

Objective: find a policy π maximising expected return:

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} \gamma^t R_t\right]$$

Symbol	Meaning
N	number of active UAV agents
$\mathcal{S}$	global system state space
$\mathcal{A}_i$	action space per UAV
$\mathcal{O}_i$	local observation per UAV
$\mathcal{T}$	state transition dynamics

Global State s_t

The full system snapshot at time t:

$$s_t = \Big\{ (p_i,\ v_i,\ b_i,\ m_i)_{i=1}^{N},\ \ (q_d)_{d=1}^{M},\ \ Q_t,\ \ F_t \Big\}$$

Symbol	Meaning	Range
p_i	position of UAV i	$\mathbb{R}^3$
v_i	velocity of UAV i	$\|v_i\| \le v_{\max}$
b_i	normalised battery level	$[0, 1]$
m_i	operating mode	$\{$serve, transit, charge$\}$
q_d	position of sensor d	fixed
Q_t	charging pad occupancy	$\{0, 1\}^S$
F_t	which UAVs have failed	$\{0, 1\}^N$

Key

The **centralised critic** sees all of s_t during **training**. Each UAV only sees its **local**

Local Observation o_i

Each UAV sees only what is nearby:

$$o_i = \big[\underbrace{p_i,\ v_i,\ b_i}_{\text{own state}},\ \underbrace{\mathcal{N}_i}_{\text{neighbours}},\ \underbrace{C_i^{\text{local}}}_{\text{local coverage}},\ \underbrace{\text{risk}_i}_{\text{cut-vertex?}},\ \underbrace{t_i^{\text{empty}}}_{\text{time-to-empty}},\ \underbrace{E_i^{\text{station}}}_{\text{energy to pad}}\big]$$

Symbol	Meaning
$\mathcal{N}_i$	indices of UAVs within R_{comm}
C_i^{local}	number of local sensors covered
risk_i	$= 1$ if removing i disconnects G_t
t_i^{empty}	seconds until battery $\leq$ return threshold
E_i^{station}	energy needed to reach nearest pad

Decentralised execution

At deployment: $a_i \sim \pi_\theta(a_i \mid o_i)$ — no global state needed.

Why Not a Central Controller?

Centralised action space grows exponentially:

$$|\mathcal{A}| = A^N$$

N	4	6	10	20
$\lvert\mathcal{A}\rvert$	$5^4 = 625$	$5^6 = 15625$	$\sim 10^7$	$\sim 10^{14}$

Solution: factorised policy — each UAV acts independently:

$$a_i \sim \pi_\theta(a_i \mid o_i) \qquad \Longrightarrow \qquad |\mathcal{A}|_{\text{eff}} = A \ \text{ per agent}$$

Trade-off

Decentralised is tractable, but each UAV must infer coordination from its local observation o_i alone.

Action Space $\mathcal{A}_i$

Each UAV picks one action per time step:

$$\mathcal{A}_i = \{\text{Hover},\ +x,\ -x,\ +y,\ -y,\ \text{Charge}\}, \quad |\mathcal{A}_i| = 6$$

Movement:

$$p_i(t+1) = p_i(t) + v_i(t)\,\Delta t, \quad \|v_i\| \le v_{\max}, \quad p_i \in \Omega$$

Charge is valid only when:

$$\|p_i - p_s^{\text{pad}}\| \le d_{\text{charge}} \quad \text{and} \quad Q_t[s] = 1 \text{ (pad is free)}$$

Otherwise the action defaults to *Hover*.

K-Coverage Requirement

Rule: every sensor d must have $\geq K$ drones above it.

Binary coverage (is UAV i covering sensor d?):

$$x_{id}(t) = \mathbf{1}\big[\, \|p_i(t) - q_d\| \leq R_{\text{cov}} \,\big]$$

Count per sensor:

$$n_d(t) = \sum_{i=1}^{N} x_{id}(t)$$

Coverage score:

$$C(t) = \frac{1}{M} \sum_{d=1}^{M} \min\left(\frac{n_d(t)}{K},\ 1 \right)$$

Under-coverage rate (to minimise):

$$\frac{1}{M} \sum_{d=1}^{M}$$

Connectivity

Drones form a **communication graph**:

$$G_t = (V, E_t), \quad E_t = \{(i,j) : \|p_i - p_j\| \le R_{\text{comm}}\}$$

Connectivity risk (is UAV i a cut-vertex?):

$$\text{risk}_i(t) = \mathbf{1}[\, G_t^{-i} \text{ is disconnected}\,]$$

Why this matters

A drone that is the *only link* between two parts of the swarm should **not** leave for charging without a replacement.

Desired property: G_t is connected $\forall\, t$.

Battery Model

Three sources of energy drain:

$$b_i(t+\Delta t) = b_i(t) \underbrace{- \frac{P_{\text{hover}} + P_{\text{move}} + P_{\text{comm}}}{B_i^{\max}} \Delta t}_{\text{drain}} + \underbrace{\Delta b_i^{\text{charge}}}_{\text{recharge (at pad)}}$$

Return-to-charge condition:

$$b_i(t) \cdot B_i^{\max} \;\leq\; E_i^{\text{req}} = E_i^{\text{station}} + E_i^{\text{return}} + E^{\text{reserve}}$$

Symbol	Meaning
P_{hover}	power consumed hovering (W)
P_{move}	power consumed moving (W)
P_{comm}	power consumed communicating (W)
$B_i^{\max}$	total battery capacity (J)
E_i^{station}	energy to reach nearest pad

Charging Coordination

Multiple drones may need to charge, but pads are limited. We solve a **min-cost assignment**.

Decision: $y_{is} \in \{0, 1\}$ — assign UAV i to pad s?

Cost:

$$c_{is} = \alpha\, d_{is} + \beta\, q_s + \mu\, u_i$$

Optimisation:

$$\min_{y} \sum_{i} \sum_{s} c_{is}\, y_{is}$$

Constraints:

$$\sum_{s} y_{is} \leq 1 \;\; \forall\, i \qquad \text{(each UAV} \leq 1 \text{ pad)}, \qquad \sum_{i} y_{is} \leq 1 \;\; \forall\, s \qquad \text{(each pad} \leq 1 \text{ UAV)}$$

Symbol Meaning

Multi-Objective Function

$$\min_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1}\Big(\underbrace{\lambda_O\, U(t)}_{\text{under-coverage}} + \underbrace{\lambda_E\, E(t)}_{\text{energy waste}} + \underbrace{\lambda_D\, D(t)}_{\text{dead time}} + \underbrace{\lambda_W\, W(t)}_{\text{wasted moves}} + \underbrace{\lambda_R\, R(t)}_{\text{recovery}} + \underbrace{\lambda_C\, \mathrm{Crit}(t)}_{\text{cut-vertex}}\Big)\right]$$

Term	Definition	Penalises
$U(t)$	$\frac{1}{M}\sum_d \mathbf{1}[n_d < K]$	sensors below K-coverage
$E(t)$	fraction of flights w/ low battery	excessive energy drain
$D(t)$	fraction of time degraded	time after a failure
$W(t)$	moves that don't help coverage	wasteful repositioning
$R(t)$	$\min(t_{\text{recovered}}, t_{\text{cap}})$	time to restore coverage
$\mathrm{Crit}(t)$	fraction of active cut-vertices	connectivity fragility

Safety Constraints

At every time step t, for all UAVs i:

1. **Speed:** $\|v_i(t)\| \leq v_{\max}$
2. **Workspace:** $p_i(t) \in \Omega \subset \mathbb{R}^3$
3. **Collision avoidance:** $\|p_i(t) - p_j(t)\| \geq d_{\text{safe}} \quad \forall j \neq i$

Hard vs Soft

These are **hard constraints**. Violated states are prevented (action masking) or heavily penalised in the reward.

Failure & Recovery

Recovery time: how long until coverage is restored after a failure at t_f:

$$\tau_f = \inf\{\tau \geq 0 : n_d(t_f + \tau) \geq K \ \ \forall\, d \in \mathcal{M}\}$$

Requirement:

$$\tau_f \leq \tau_{\max}$$

Reserve UAV Strategy

Keep 1 drone in reserve with $b_i > b_{\text{threshold}}$.

When a drone fails, the reserve **immediately repositions** to fill the gap, reducing τ_f.

Activation: reserve takes off when F_t changes (new failure detected).

Learning Algorithm: MAPPO

Multi-Agent Proximal Policy Optimisation.

Actor (per UAV): $a_i \sim \pi_\theta(a_i \mid o_i)$

Critic (centralised, training only): $V_\phi(s_t)$

Actor loss (PPO clipped):

$$\mathcal{L}^{\text{actor}}(\theta) = -\mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta), 1-\epsilon, 1+\epsilon\big)\,\hat{A}_t\Big)\Big]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{\text{old}}}(a_t|o_t)}$, $\hat{A}_t = R_t - V_\phi(s_t)$ (GAE).

Critic loss:

$$\mathcal{L}^{\text{critic}}(\phi) = \mathbb{E}_t\Big[\big(R_t - V_\phi(s_t)\big)^2\Big]$$

Training: centralised critic sees s_t. **Execution:** each UAV acts on o_i alone.

Simulation Setup

Parameter	Symbol	Value
Airspace	Ω	$100 \times 100 \times 30$ m^3
Active UAVs	N	5
Reserve UAVs	—	1
IoT sensors	M	20
Coverage radius	R_{cov}	15 m
Comm radius	R_{comm}	40 m
Min coverage	K	2
Sim step	Δt	1 s
Max velocity	$v_{\max}$	5 m/s
Safe distance	d_{safe}	5 m
Battery capacity	$B^{\max}$	30 000 J
Hover power	P_{hover}	50 W
Move power	P_{move}	75 W
Comm power	P_{comm}	10 W

Questions?